From Verification to Forecasting: Monitoring Official Crime Trends in Bangladesh

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***Abstract*—Bangladesh Police Headquarters (PHQ) publishes monthly crime statistics. We use these official numbers to build a forecasting and monitoring system that police planners and the public can trust. Our work has three honest goals: the data must be checked against the original government documents, the accuracy must be tested in a fair way (no peeking at the future), and the whole thing must keep working month after month as new data arrives. The dataset covers January 2019 to April 2026 (88 months, 17 police units, 15 crime types). Every month is traced to the official signed PHQ statements and passed a five-step audit against 22 source documents: all 88 months reconcile internally, the six complete years are cross-checked against PHQ's independently printed annual pages, and a 429-cell manual re-audit of the most recent window matched the source with zero mismatches. We trained seven standard forecasting models and an average (ensemble) of them on data up to April 2025, then forecast 1, 3, 6, and 12 months ahead and compared the forecast with the real numbers that PHQ released later. In this single-origin back-test of national Total Cases, the forecast was 98.9% accurate at 1 month, 97.8% at 3 months, 92.8% at 6 months, and 86.1% at 12 months; a separate rolling-origin validation, averaged over the stable crime categories, ranged from 94.7% (1 month) to 89.9% (12 months). The two are measured differently and reported separately. The ensemble was the most reliable model. We add a plain-language explanation layer that shows which past months the model leans on and which crime types are hard to forecast, and we ship a working monitoring application that ingests each new month, re-checks the previous forecast, and updates the models. A Transformer model was tested but did not help at this data size. The forecasting numbers are a planning scenario only, not a verified prediction. The system forecasts *reported* crime counts; it does not measure hidden crime and does not claim to explain why crime rises or falls.**

***Index Terms*—reported-crime forecasting, rolling-origin validation, backtesting, MAPE, time-series, Bangladesh, Police Headquarters, source verification, operational monitoring, explainability**

I. Introduction

Crime statistics matter for a country. They guide where police are sent, how budgets are spent, and how leaders plan for public safety. In Bangladesh, Police Headquarters (PHQ) releases a crime report every month. If those monthly numbers can be forecast reliably, planners can prepare ahead instead of only reacting after the fact.

This paper is written so that it can be read by more than just machine-learning researchers. A police officer, a journalist, a student, or any citizen who cares about the country should be able to follow what we did and why it can be trusted. We therefore explain the technical words in plain language and keep the claims modest and honest.

We are careful about one thing from the start. We forecast *reported* crime, the crimes that were recorded by police and printed in the official PHQ statements. Reported crime is only a part of all real crime, because not every crime is reported. So this is not a tool that predicts true crime, and it is not a tool that predicts who will commit a crime. It is a tool that forecasts the official monthly counts and watches how those counts move over time.

A. Why this work can be trusted

The main point of this paper is not one big accuracy number. The stronger point is a pipeline that is honest end to end:

- **Verified data.** Every number can be traced back to a signed PHQ document.
- **Fair testing.** Accuracy is measured without ever letting the model see the future (we explain how in Section IV-A), and then confirmed by comparing an old forecast against months that were released only later.
- **Keeps working.** The forecasting logic is wrapped in a monitoring application that updates itself every month.

We avoid hype. We do not call this a "crime prediction engine," and we make no accuracy claim for the scenario projection.

B. Contributions

This paper makes four contributions:

1) **A verified PHQ dataset.** A reported-crime dataset for January 2019 to April 2026, traced to the official signed PHQ statements and checked with a five-step audit against 22 source documents: all 88 months reconcile internally, the six complete years (2019–2024) are cross-checked against PHQ's independently printed annual summary pages, and a 429-cell manual re-audit of the recent window matched the source with zero mismatches.
2) **Fair, leak-free testing.** A rolling-origin validation and backtesting framework. We train only on data up to April 2025, forecast 1, 3, 6, and 12 months ahead, and compare against the real PHQ numbers released afterward.

3) **Plain-language explainability.** A diagnostic layer that shows which inputs the model trusts and which crime types are easy or hard to forecast.
4) **A working monitoring application.** Deployable software that adds each new month, automatically re-checks the previous forecast, retrains, and watches for drift.

II. Related Work

A. Crime Forecasting in Bangladesh

A few studies have applied machine learning to Bangladesh crime data. Awal et al. [1] used linear regression to forecast crime trends from Bangladesh Police records. Biswas and Basak [2] extended this at an IEEE conference with polynomial and random-forest regression on PHQ website data through 2018. Nahid [3] applied Random Forest, MLP, and Decision Tree classifiers to PHQ data for five police units. Shohan et al. [4] worked on a different problem, classifying incident-level crime from newspaper articles.

We looked closely at what these papers documented. Biswas and Basak [2] describe training their regression models on a training set and then forecasting on a separate test set and comparing with the actual values, that is, a single fixed train–test split rather than a time-series protocol that rolls forward month by month. Across these studies, after researching the published papers, *we have not seen documented evidence* of three things that our work adds:

- a source-verification audit that checks each number against the original PHQ documents;
- post-release back-testing, where a forecast made at a fixed past date is later compared against months that were published only afterward;
- an explainability layer that shows which crime types drive the forecast error.

We state this carefully. We are not claiming these studies did nothing useful; we are saying that, in their published text, we did not find these specific safeguards. Our contribution is to add them.

B. Forecasting Methods and Validation

Testing a forecast on data it has already seen gives a falsely high score. Tashman [5] reviews out-of-sample testing and recommends rolling-origin evaluation instead of a single split. Bergmeir and Benítez [6] show that rolling-origin protocols give more reliable error estimates than random splits, which can leak future information. For measuring error, Hyndman and Koehler [7] analyse percentage- and scale-based measures and propose the mean absolute scaled error (MASE); we report MAPE, WAPE, sMAPE, MAE, RMSE, and a seasonal MASE. Classical models remain strong on short monthly data: the ARIMA/SARIMA family [8], exponential smoothing (ETS) [9], and the Theta method [10], whose strength on monthly series was shown in the M3 competition [11]. Gradient-boosted trees such as LightGBM [12] give a machine-learning baseline; we use statsmodels [13] for the classical models. Transformer models such as the Temporal Fusion Transformer [14] and PatchTST [15] do well on large datasets; we test a lightweight Transformer only to check feasibility, because our history is short.

C. Explainability Methods

Tree models such as LightGBM [12] report a built-in importance score, which together with permutation importance [16] and SHAP [17] forms a standard toolkit. Local methods such as LIME [18] are also common. Surveys of machine-learning crime forecasting [19] stress reported-crime data and public-safety uses, which is exactly our setting. We combine these tools into a simple explanation layer.

III. Dataset and Source Verification

The data comes from the official monthly reported-crime statements released by Police Headquarters, Bangladesh [20]. It covers January 2019 to April 2026 (88 months in a row, with no missing months). There are 17 reporting units: eight metropolitan police units (DMP, CMP, KMP, RMP, BMP, SMP, RPMP, GMP), eight ranges (Dhaka, Mymensingh, Chittagong, Sylhet, Khulna, Barishal, Rajshahi, Rangpur), and the Railway Range. Together these cover the whole country, so adding them up gives a national total. Each month lists 15 crime types: Dacoity, Robbery, Murder, Speedy Trial, Riot, Woman & Child Repression, Kidnapping, Police Assault, Burglary, Theft, Other Cases, and four recovery types (Arms Act, Explosive Act, Narcotics, Smuggling). “Total Cases” is the sum of all 15 types, matching the official total printed in the statements.

A. Five-Step Source Verification

A forecast is only as good as the data behind it. We checked the dataset in five steps, in plain terms:

1) **Do the numbers add up?** We used spreadsheet formulas across 1,584 unit-month rows to confirm the 15 crime types add up to the printed Total Cases for every unit and month.
2) **Does each number match the original document?** We opened 22 official PHQ statement PDFs and compared the numbers in our spreadsheet against them, cell by cell.
3) **Where did each number come from?** We recorded the source (which document, which month) for every figure, so anyone can trace it back.
4) **Do the yearly totals match?** PHQ also publishes yearly summaries. We added up our monthly numbers for each year and checked them against the published yearly totals (results below).
5) **Did our past forecast match reality?** We made a forecast at a fixed past date and later compared it with the months PHQ released afterward (Section V).

Every one of the 88 months is printed in one of the 22 signed PHQ source documents (recent months in their own monthly statements, earlier years in PHQ’s consolidated yearly statements) and was checked against it. All 88 months reconcile internally: the 15 crime types sum to the printed Total Cases for every unit and month, across 1,584 unit-month rows. For the six complete years (2019 to 2024), the monthly figures

TABLE I
DATASET AND PHQ SOURCE-CHECK SUMMARY.

Property	Value
Source-verified coverage	2019-01 to 2026-04 (88 months)
Forecasting data	2019-01 to 2026-04 (88 months)
Training cutoff for back-test	April 2025 (76 months)
Frequency	monthly, no gaps
Reporting units	17 (covers the whole country)
Crime types per month	15 (+ Total Cases)
Source documents checked	22 official PHQ statements
Internal consistency	all 88 months reconcile (zero errors)
Annual cross-check (2019–2024)	exact (1 documented source error)
Recent-window cell re-audit	429 of 429 matched (zero)

were additionally cross-checked against PHQ's independently printed annual summary pages; 2019 to 2023 match exactly on all 17 columns, while 2024 matches on 16 of 17 columns and the grand total, the sole exception being a documented error on PHQ's own 2024 annual page (an RPMP Robbery/Murder mislabel that leaves every total unaffected). The most recent window (2025 to April 2026), which has no annual summary page yet, was re-audited cell by cell: three months as full visual checks against the signed scans and thirteen months number by number (unit totals, national totals, category totals, headers, and unit order), giving 429 of 429 matching comparisons with zero mismatches. Any discrepancies found during transcription (source typographical errors and a few ambiguous scanned digits) are catalogued and resolved in the dataset's audit-findings log, with none outstanding. Table I summarizes the dataset.

IV. METHODOLOGY

A. Terminology

A few technical words appear in this paper. Here is what they mean, explained simply:

- **Forecast horizon.** How far ahead we predict. We use horizons of 1, 3, 6, and 12 months; a 3-month horizon means predicting the value three months into the future.
- **Backtesting (looking back to check).** We pretend it is a past date (say April 2025) and make a forecast using only the data available up to then. Then we wait for the real months to arrive and check how close the forecast was. It is like sealing a prediction in an envelope and opening it later to see if it was right.
- **Rolling-origin validation.** Instead of testing once, we test many times. We train on the first stretch of months, forecast the next few, and check. Then we move the starting point forward and repeat. We keep "rolling" forward, always testing on months the model has not learned from. This gives a fair, repeated measure of accuracy.
- **Leak-free (no peeking at the future).** When we test a forecast, the model is never allowed to use any data from after the forecast date. Letting it peek would be cheating and would make the score look better than it really is. We never peek.
- **Practical accuracy.** For easy reading we report accuracy as $100 - \text{MAPE}$, where MAPE is the average percentage error. So a 7% average error reads as 93% practical accuracy. This is just a readable label; MAPE and MASE are our real error measures.

B. Forecasting Models

We use a transparent set of seven forecasting models, plus an ensemble (a simple average of ETS, Theta, ARIMA, SARIMA, and LightGBM). Every model is standard and easy to inspect:

- **Naive** and **Seasonal Naive**: simple baselines (repeat the last value, or the value from the same month last year).
- **ETS**, **Theta**, **ARIMA**, **SARIMA**: classical time-series models that capture trend and yearly seasonality.
- **LightGBM**: a gradient-boosted tree using past months as features.
- **Ensemble**: the average of ETS, Theta, ARIMA, SARIMA, and LightGBM. Averaging reduces the risk of any single model going wrong.

If a model ever produces a broken or unrealistic number (non-finite, negative, or implausibly large), the system automatically falls back to the Seasonal Naive forecast and records the event in a persistent audit log. This guard runs identically in the analysis notebook and in the monitoring software, so a single bad fit can never silently spoil the results.

C. Validation and Back-Testing Protocol

We tested in two ways that support each other:

- **Rolling-origin validation** over the full data, at horizons of 1, 3, 6, and 12 months, using an expanding training window.
- **A fixed back-test:** we trained only on January 2019 to April 2025, then forecast 1, 3, 6, and 12 months ahead, landing on May 2025, July 2025, October 2025, and April 2026. We compared each forecast with the real, later-released PHQ number for that month.

We group the crime types into *stable* types (Total Cases, Murder, Theft, Woman & Child Repression, Robbery), which have large, steady monthly counts, and *hard* types (Narcotics recovery, Kidnapping, Dacoity), whose counts are small and jumpy. Our headline accuracy uses the stable types only, and we are open about why (Section VI).

V. RESULTS

We report two things separately and never mix them: (A) the fixed back-test of the April 2025 forecast, and (B) the rolling-origin validation over all the data.

A. Post-Release Back-Test of the April 2025 Forecast

This is the strongest evidence in the paper. We made a forecast in April 2025, using *only* data up to that point, and then checked it against the real PHQ numbers that came out

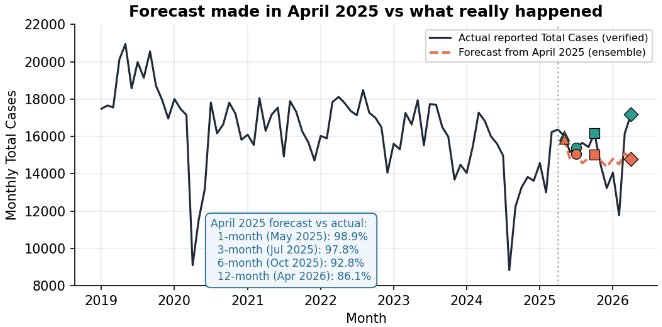

Fig. 1. The forecast made in April 2025 versus the real PHQ numbers released later (national Total Cases, single origin). The forecast tracks reality closely: 98.9% accurate at 1 month, 97.8% at 3 months, 92.8% at 6 months, and 86.1% at 12 months. This is a fair test: the model never saw any data after April 2025.

later. This is a single forecast origin for one series, national Total Cases. The ensemble forecast was:

- **98.9%** accurate 1 month ahead (May 2025),
- **97.8%** accurate 3 months ahead (July 2025),
- **92.8%** accurate 6 months ahead (October 2025),
- **86.1%** accurate 12 months ahead (April 2026).

All four figures are for national Total Cases at this single April 2025 origin, not the rolling-origin averages over the five stable crime types in Table II (where the 12-month figure is 89.9%). A single national total swings more than an average over stable series, so its long-horizon accuracy is naturally a little lower; the two are measured differently and reported separately. Fig. 1 shows the full picture: the dark line is the real, verified history; the dashed line is the April 2025 forecast; and the marked points are the four checks. The forecast stays close to reality, and accuracy is naturally highest at the short horizon and slowly drops as we look further ahead, exactly what we expect from an honest forecast. This same checking step is built into the monitoring loop (Section VII): every new month becomes a fresh test of the last forecast.

B. Rolling-Origin Validation

The repeated rolling-origin test agrees with the back-test. Unlike the single-origin back-test above, these numbers are averaged over many forecast origins and over the five stable crime types, so they are not expected to match the back-test figure for any single month; the two tests are complementary, not contradictory. For the stable crime types, the average error stays low at every horizon, well under the 18% line (82% accuracy) that we treat as the minimum bar (Table II, Fig. 2). Using a "furthest horizon that still passes" rule, the 12-month horizon (89.9% accuracy) is our headline, while the 1-month horizon is the most accurate at 94.7%.

The ensemble was the most reliable model overall, with the lowest average error and beating the Seasonal Naive baseline in 75% of the test cases (Table III, Fig. 3). A MASE below 1.0 means a model beats the seasonal baseline on average; the ensemble reaches 0.577, and every classical model here is below 1.0. LightGBM was not competitive on this short

TABLE II
ROLLING-ORIGIN ACCURACY FOR STABLE CRIME TYPES, BY HORIZON.

Horizon	MAPE (%)	Practical acc. (%)	MASE	Status
1	5.27	94.7	0.346	pass
3	7.34	92.7	0.500	pass
6	8.33	91.7	0.544	pass
12	10.13	89.9	0.674	pass (headline)

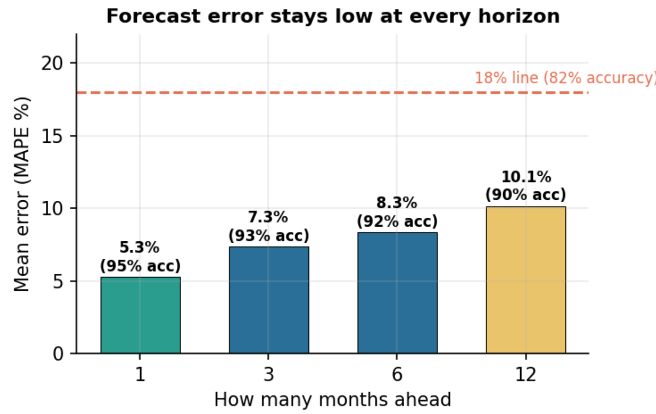

Fig. 2. Average error for stable crime types at each horizon. Every bar stays below the 18% line (82% accuracy), which is why we can stand behind a 12-month forecast, not just a 1-month one.

monthly history, which is expected: tree models need more data than we have.

C. Scenario Projection to December 2027

Using all 88 verified months, we extend the ensemble forecast to December 2027 (Fig. 4). **These numbers are a planning scenario, not a verified prediction.** We attach no accuracy claim to them. They are useful for rough planning, and the shaded band is a reminder that the real values will differ.

For concreteness, the scenario keeps national monthly Total Cases roughly in the 14,600 to 16,600 range through 2027, ending near 14,600 in December 2027, for an implied 2027 total of about 184,000 cases (Table IV). These remain planning figures with no accuracy claim.

VI. EXPLAINABILITY AND ERROR ANALYSIS

A forecast people can trust should also be a forecast people can understand. We add a simple explanation layer. It does not change the models or the accuracy; it only shows *what the model is paying attention to* and *where it struggles*.

A. Feature Importance

We trained a LightGBM model on the same past-month features the pipeline uses and read both its built-in importance and a model-agnostic permutation check [16]. Both point the same way (Fig. 5):

- the **most recent month** matters most;

TABLE III
MODEL COMPARISON OVER STABLE CRIME TYPES (LOWER ERROR IS BETTER).

Model	Mean MAPE (%)	Mean MASE	Beats Seasonal Naive (%)
Ensemble	**8.96**	**0.577**	**75**
ETS	9.64	0.633	55
Seasonal Naive	9.84	0.636	n/a
Theta	10.19	0.670	55
SARIMA	10.23	0.672	20
ARIMA	10.63	0.702	45
Naive	12.35	0.819	15
LightGBM	14.82	0.890	25

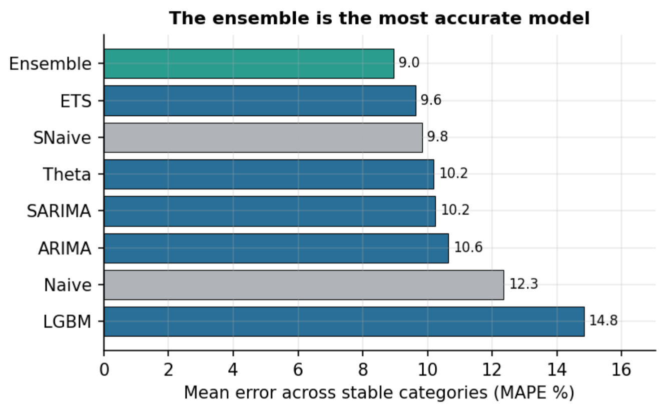

Fig. 3. Average error by model across stable crime types. The ensemble (green) is the most accurate. Tree-based LightGBM trails because the monthly history is short.

- the **recent ups and downs** (how bumpy the last few months were) matter next;
- the **month of the year** matters too, which fits the yearly pattern in crime data.

Because two different methods agree, we can be confident the model is using real, sensible signals (recent level and seasonality), not random noise.

B. Error by Crime Category

We also measured the error for each crime type (Fig. 6). Two clear patterns appear:

- **Stable types** (Theft at 6.4% error, Total Cases at 7.0%, Murder at 8.6%) are easy to forecast because their monthly counts are large and steady.
- **Hard types** (Kidnapping at 22%, Dacoity at 18%) are difficult because their counts are small, so even a swing of a few cases looks like a big percentage error.

This is exactly why our headline accuracy uses the stable types only and openly keeps the hard types separate. Hiding the hard types would be dishonest; reporting them as headline accuracy would be misleading. We do neither.

C. Scope of the Explainability Layer

The explanation layer is a diagnostic, not a cause. It shows what the model leans on; it does not prove that crime in the real world works that way. It shows where forecasts miss; it does not explain why crime itself went up or down. It changes none of the accuracy numbers above.

TABLE IV
SCENARIO PROJECTION FOR NATIONAL TOTAL CASES (PLANNING ONLY; NO VALIDATED ACCURACY CLAIM). HORIZONS ARE MEASURED FROM THE LAST OBSERVED MONTH, APRIL 2026.

Horizon	Target month	Projected Total Cases
+1 month	May 2026	16,639
+3 months	July 2026	16,118
+6 months	October 2026	15,352
+12 months	April 2027	16,142
+20 months	December 2027	14,587

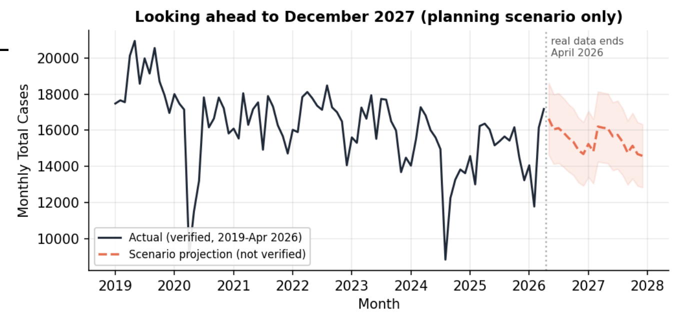

Fig. 4. Verified history (to April 2026) with a planning-only projection to December 2027. The vertical line marks where real data ends. No accuracy is claimed for the projected part.

VII. OPERATIONAL MONITORING SYSTEM

A research result that sits in a notebook helps no one. We built a working application so the framework can run as a monthly routine inside a police or planning office. Fig. 7 shows the loop. In plain terms, the software does this:

- **Adds a new month** from a CSV or Excel file, and checks it automatically (missing units, missing crime types, or impossible values are flagged before anything is saved).
- **Re-checks the last forecast** against the new real numbers, so accuracy is tracked as it happens.
- **Retrains the models** on the updated data and produces the next forecast.
- **Watches for drift**: if accuracy starts slipping, it raises a warning and recommends recalibration.
- **Shows the explanation views** (which inputs matter, which crime types are hard).
- **Controls access by role** (administrator, operator, reviewer, viewer), so data entry and approval are separated.

The application keeps a full audit trail (who added or approved what, and when), stores every forecast so it can be scored later, and writes any model fallback to that same persistent audit log. It was checked with an automated smoke-test suite of 18 checks across seven groups: seed-data integrity, duplicate-month safety, forecasting completion, recalibration persistence, forecast-vs-actual lookup, restart-safe database reads,

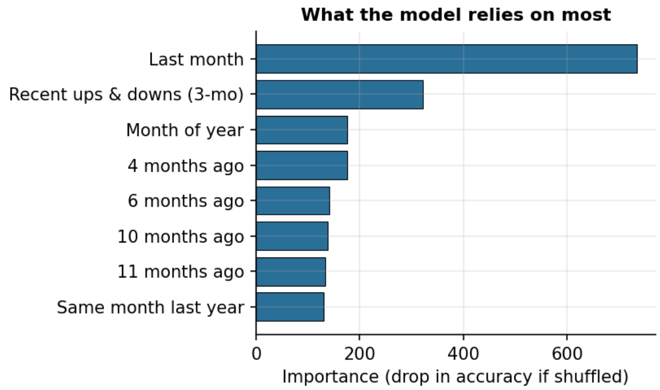

Fig. 5. What the model relies on most, measured by how much accuracy drops when each input is shuffled. The last month and recent movement lead, with month-of-year close behind, which is sensible, not noise.

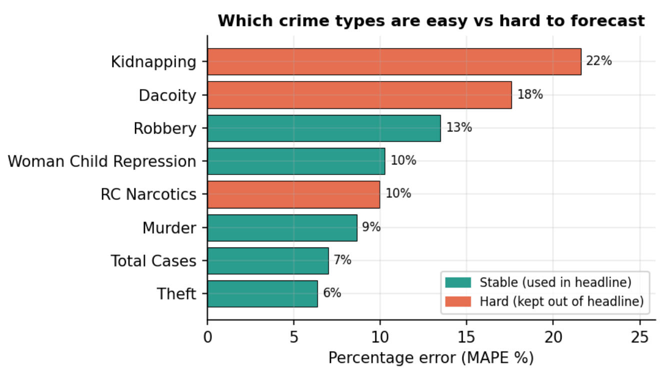

Fig. 6. Percentage error by crime type. Green types are stable and used in the headline; orange types are hard (small, jumpy counts) and kept separate. The split is reported openly.

and role-based authentication; all 18 pass. It is straightforward to deploy on a single office machine or a small server.

VIII. Limitations

We want readers to know exactly what this system is and is not:

- It forecasts **reported** crime (what police recorded), not all real crime. Unreported crime is outside its reach.
- It does **not** predict individuals or specific locations. It forecasts monthly counts.
- The monthly history is **short** (88 months). This is why simple, classical models and their average beat heavier deep-learning models here.
- Accuracy **drops with distance**: short-term forecasts are stronger than long-term ones.
- The future projection is a **planning scenario**, not a verified forecast.
- Forecasts can be disturbed by **sudden shocks** (for example, the large drop in early 2020 during the pandemic, which is visible in our data). No model can foresee such events.

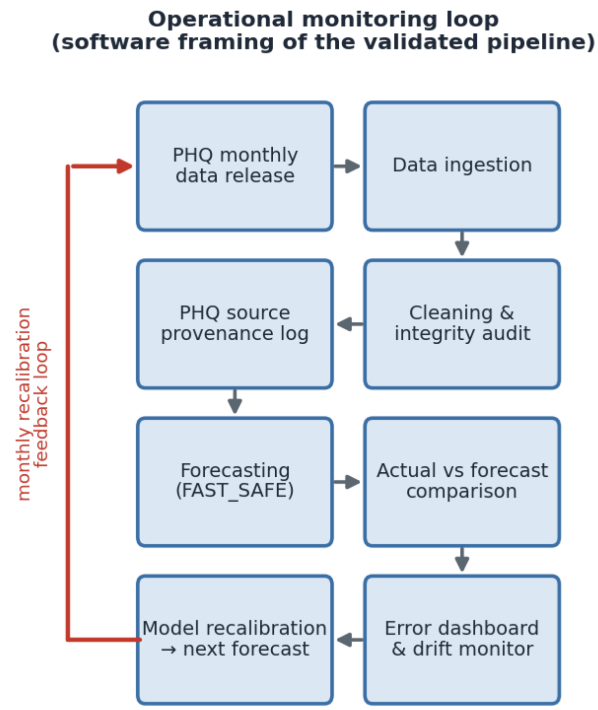

Fig. 7. The monthly monitoring loop. Each new PHQ release is added and checked, the previous forecast is scored against it, drift is watched, the models are retrained, and the next forecast is produced, then the cycle repeats.

IX. Future Work

- Add external covariates (for example population or economic indicators) under the same source-checking and back-testing rules.
- Add district-level forecasting under the same checking and testing rules.
- Explore privacy-preserving federated learning [21], [22] so multiple police units could improve a shared model without sharing raw records.
- Add automatic alerts (email or dashboard) when drift crosses the warning level.

X. Conclusion

The real value of this work is not a single accuracy number; it is a trustworthy pipeline from verified government data to a forecast that has been tested fairly and runs every month. When we made a forecast in April 2025 and checked it against the months PHQ released later, it was 98.9% accurate at 1 month, 97.8% at 3 months, 92.8% at 6 months, and 86.1% at 12 months for national Total Cases. The ensemble of classical models was the most reliable. We keep the stable and hard crime types honestly separate, we explain what the model relies on in plain language, and we ship a working monitoring application so the framework can serve real planning needs. The forecasting numbers are a planning scenario only. Built and used responsibly, a tool like this can help Bangladesh plan for public safety with clear eyes and honest numbers.

Reproducibility

The verified dataset [23], the analysis notebook with all results and figures, and the monitoring application [24] are provided as public companion repositories, so every number and figure in this paper can be regenerated.

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